

Jan Kula

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📍 Sweden, EU

✂ AlbertEinstein

🐙 example

🌐 example

Education

Chalmers University of Technology, Mathematics

Gothenburg, Sweden

- Focused on mathematics, statistics, optimisation, and computational methods, complemented by advanced programming and applied problem-solving.

2020 – 2026

Experience

Sendify, Master's Thesis Student

Gothenburg, Sweden

Built data pipelines and trained machine learning models using Python, SQL, Pandas, and scikit-learn.

2025 – 2025

1 year

- Learned to visualize large-scale data, using matplotlib, and present key findings to non-technical parts of the team.
- Learned to structure complex problems and manage scope creep through clearly defined goals and continuous dialogue with company leadership and the technical team.

Intize, Treasurer, Math Mentor, and Board Member

Gothenburg, Sweden

Managed a heavily reduced annual budget while ensuring operations continued to run effectively, thereby gaining practical experience with grant applications and formal reporting to financial backers.

2023 – 2025

2 years

- As a board member, I contributed to strategic planning and the long-term development of the organization.
- Learned how to present complex problems in a simple and accessible manner.

Skills

Programming Python

Mathematics

Languages

Swedish 🇸🇪

Native

English 🇬🇧

Full professional proficiency / Fluent

Polish 🇵🇱

Professional working proficiency

Interests

Mathematics

Homelabbing

Movies

Projects

GPU-Accelerated FEM Solver; From Heat Equation to Black-Scholes - Chalmers

Nov 2025 – Jan 2026

Created a solver for the heat equation using Finite Element Methods. Used graphics cards in HPC environment to speed up calculations.

- Learned to use Python and GPU-accelerated libraries such as CuPy and Numba CUDA, alongside standard scientific computing tools like SciPy and NumPy.
- Gained an understanding of how differences in memory and thread management between the CPU and GPU affect execution, and how algorithms must be adapted and restructured to run efficiently on the GPU.
- Learned to operate inside of a HPC environment with resource allocation using Bash-scripts.

Predicting Molecular Fragmentation Using Graph-to-Graph Models - AstraZeneca

Nov 2024 – Jan 2025

In a group of three, researched ... and tried to reverse engineer an existing Graph Neural Network used for predicting molecular synthesis to fit our need of predicting molecular fragmentations.

- Used Python to adapt existing graph-based neural network for our purpose.
- Learned to quickly familiarize myself with a research-heavy field.
- Learned to work in a structured manner within a team environment.

References

Erik Öhrn - Engineering Manager - Sendify